

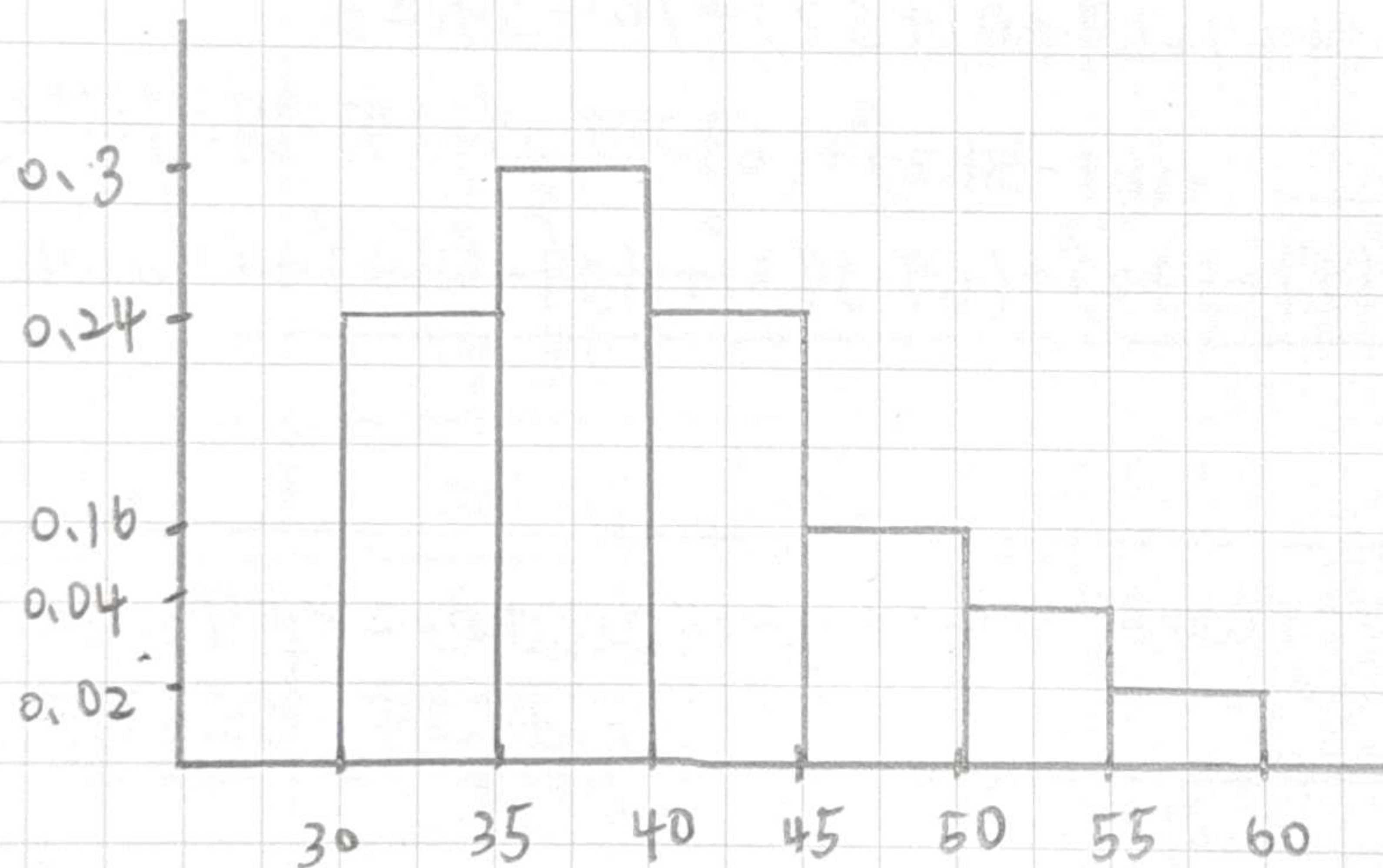
統計學作業

§ 1.4 Q2b

a.

年齡(月)	人數	relative frequency
30 to < 35	12	$12 \div 50 = 0.24$
35 to < 40	15	$15 \div 50 = 0.3$
40 to < 45	12	$12 \div 50 = 0.24$
45 to < 50	8	$8 \div 50 = 0.16$
50 to < 55	2	$2 \div 50 = 0.04$
55 to < 60	1	$1 \div 50 = 0.02$

relative frequency histogram



b.

age 35 to < 45 : $0.3 + 0.24 = 0.54$ #

c.

age < 50 : $(12 + 15 + 12 + 8) \div 50 = 94\%$ #

§ 2.1, Q2

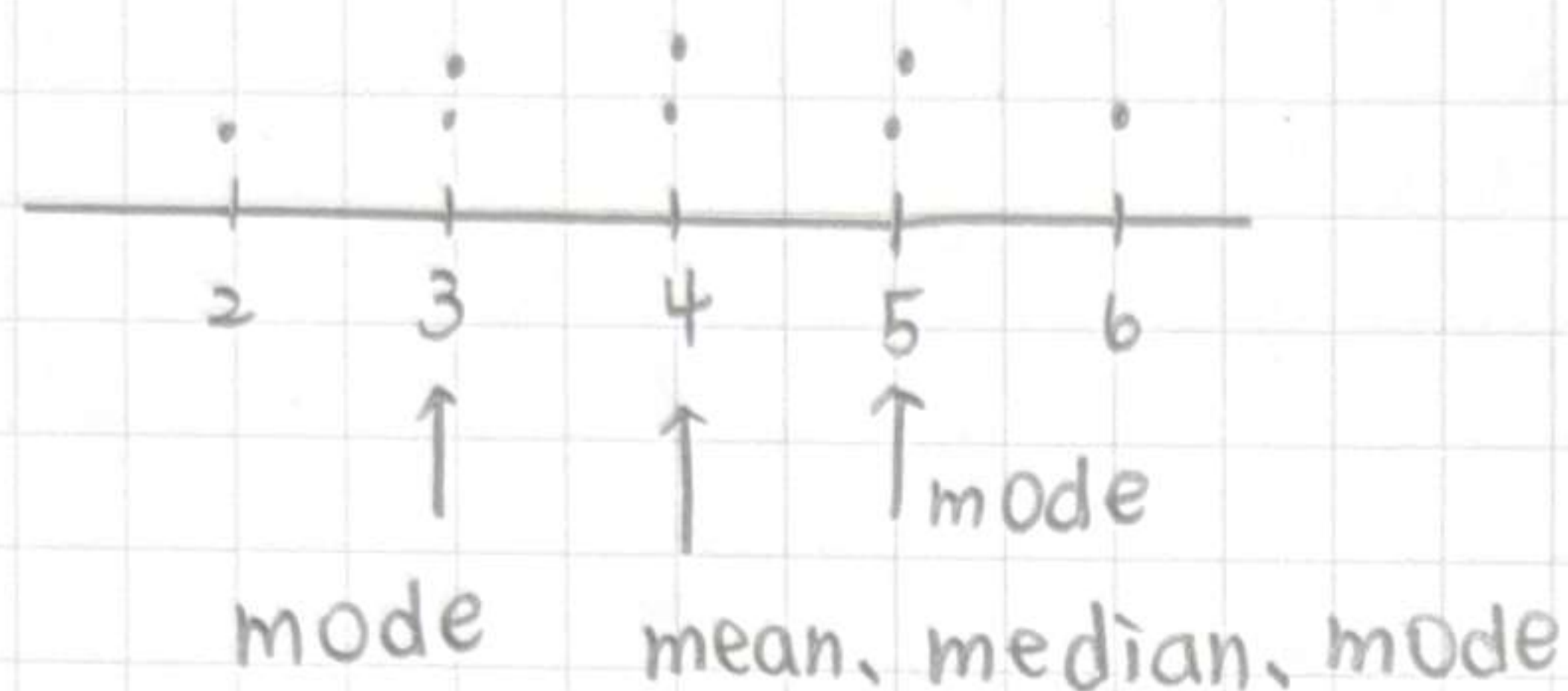
2 3 3 4 4 5 5 6

1. mean: $(3+2+5+6+4+4+3+5) \div 8 = 4 \#$

2. median: $0.5(8+1) = 4.5$, $4 + 0.5(4-4) = 4 \#$

3. mode: 3, 4, 5 #

4. Dot plot



§ 2.2, Q14

a. range: $71 - 40 = 31 \#$

b. $\bar{x}$: $(49+70+54+67+59+40+61+69+71+52) \div 10 = 59.2 \#$

$$S = \sqrt{\frac{(49-59.2)^2 + (70-59.2)^2 + (54-59.2)^2 + (67-59.2)^2 + (59-59.2)^2 + (40-59.2)^2 + (61-59.2)^2 + (69-59.2)^2 + (71-59.2)^2 + (52-59.2)^2}{10-1}}$$

$$S = \sqrt{\frac{104.04 + 116.64 + 27.04 + 60.84 + 0.04 + 368.64 + 3.24 + 96.04 + 139.24 + 50.41}{9}}$$

$S = 10.361 \#$

c. $31 \div 10.361 = 2.99 \approx 3 \#$

§2.4 Q2

1.0, 1.7, 2.0, 2.1, 2.3, 2.8, 2.9, 4.4, 5.1, 6.5, 7.8

1. mean:

$$(1.0 + 1.7 + 2.0 + 2.1 + 2.3 + 2.8 + 2.9 + 4.4 + 5.1 + 6.5 + 7.8) \div 11 = 3.51\#$$

2. standard deviation

X_i	$(X_i)^2$	
1	1	✓
1.7	2.89	✓
2.0	4	✓
2.1	4.41	✓
2.3	5.29	✓
2.8	7.84	✓
2.9	8.41	✓
4.4	19.36	✓
5.1	26.01	✓
6.5	42.25	✓
7.8	60.84	✓
SUM 38.6	182.3	

$$S^2 = \frac{182.3 - \frac{(38.6)^2}{11}}{11-1} = \frac{46.849}{10} = 4.685$$

$$S = \sqrt{4.685} = 2.164\#$$

3. Z-score

$$\text{minimum} = \frac{1 - 3.51}{2.164} = -1.160\#$$

$$\text{maximum} = \frac{7.8 - 3.51}{2.164} = 1.98\#$$

§2.4 Q20

1.7, 1.72, 5.9, 8.8, 85.40, 101.00, 118.00, 168, 180, 183, 209, 218
221, 241, 252, 264, 278, 286, 314, 315, 316, 318, 329, 397, 406
445, 481, 485

a.

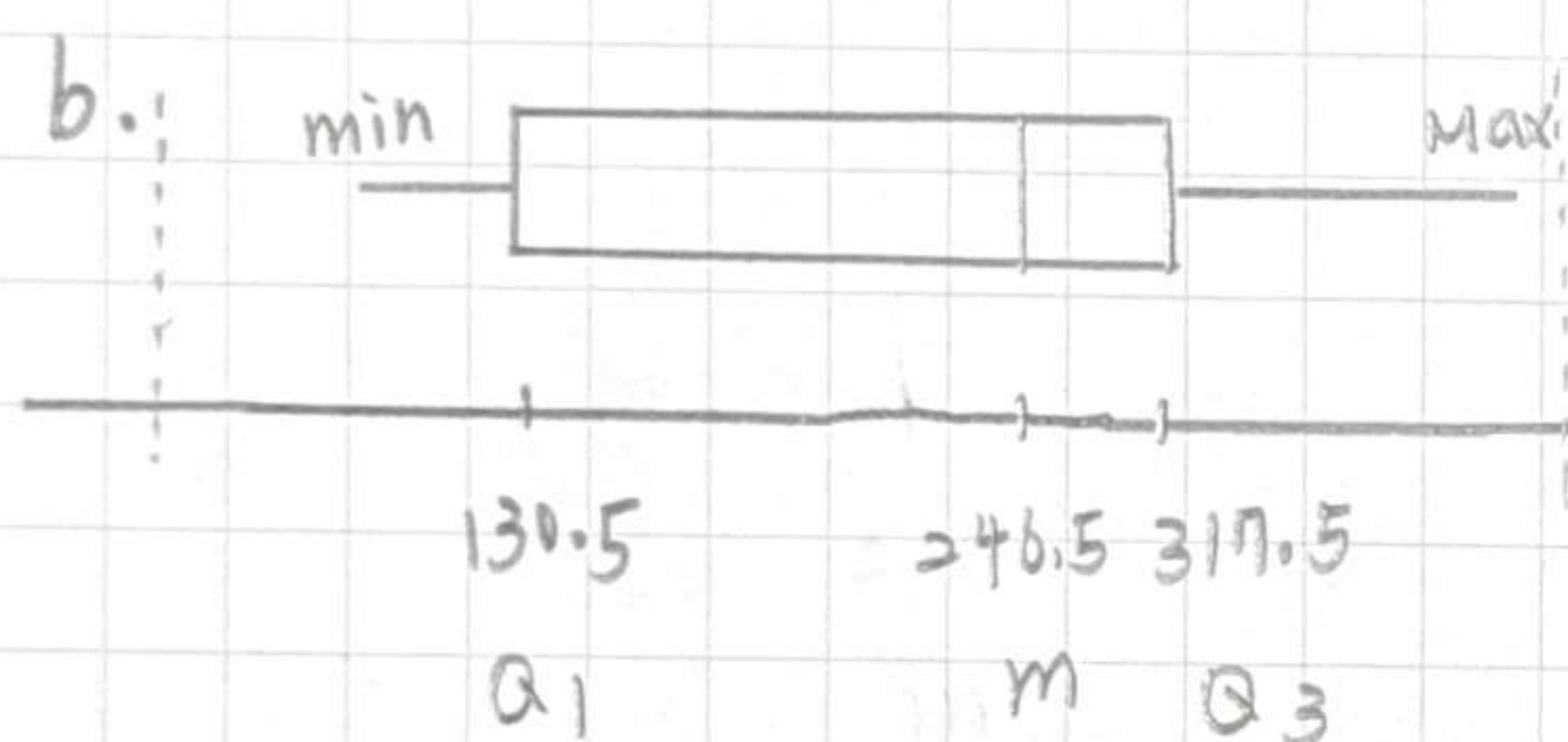
(1) mean: 1.7

(2) Max: 485

(3) Median: $0.5(28+1)=14.5$, $241 + 0.5(252-241)=246.5$

(4) $Q_1 = 0.25(28+1)=7.25$, $118 + 0.25(168-118)=130.5$

(5) $Q_3 = 0.75(28+1)=21.75$, $316 + 0.75(318-316)=317.5$



$$IQR = Q_3 - Q_1 = 187$$

$$\text{Lower fence} = Q_1 - 1.5IQR = 130.5 - 280.5 = -150$$

$$\text{Upper fence} = Q_3 + 1.5IQR = 317.5 + 280.5 = 598$$

c. 沒有 outlier

d. 可以, 年齡越小, 攝取食物的時間較少, 故體內累積的重金屬較少

§2.4 Q24 $233.97, 243.92, 247.34, 252.44, 255.40, 273.80, 286.35, 286.41, 383.68, 408.48, 446.30, 459.21$

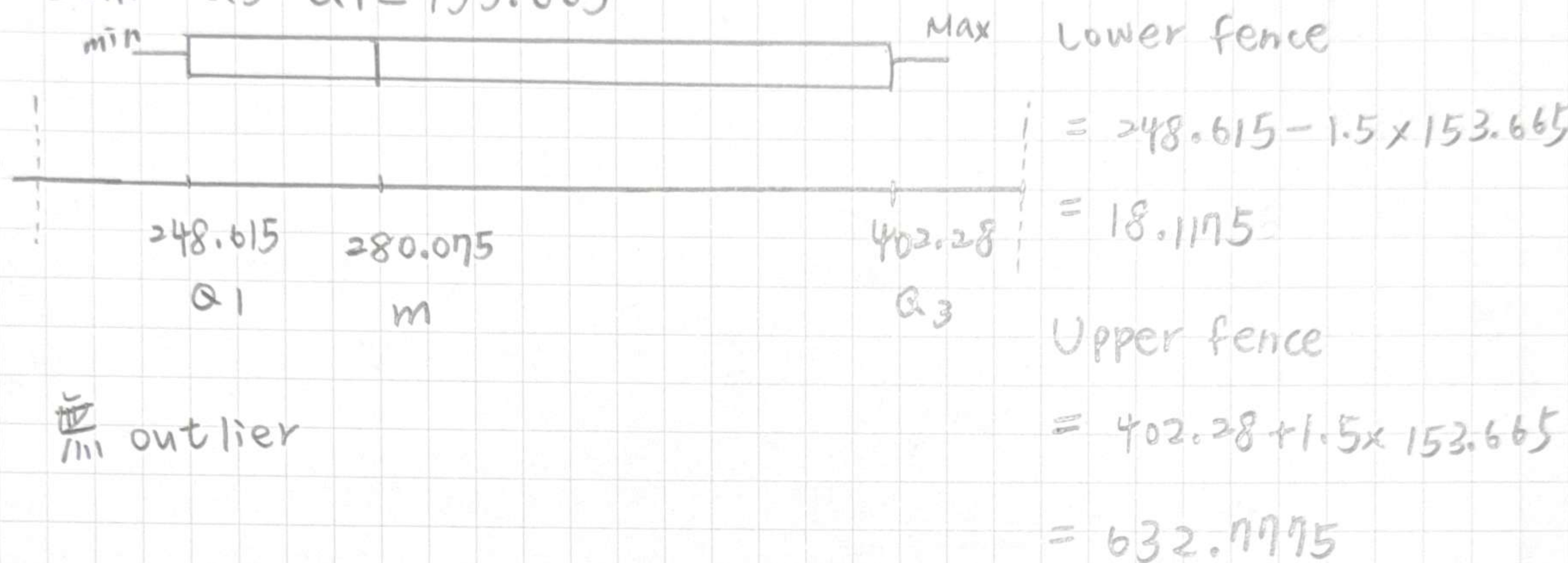
a.

$$\text{median} = 0.5(12+1) = 6.5, 273.80 + 0.5(286.35 - 273.80) = 280.075$$

$$Q1 = 0.25(12+1) = 3.25, 247.34 + 0.25(252.44 - 247.34) = 248.615$$

$$Q3 = 0.75(12+1) = 9.75, 383.68 + 0.75(408.48 - 383.68) = 402.28$$

$$IQR = Q3 - Q1 = 153.665$$



b.

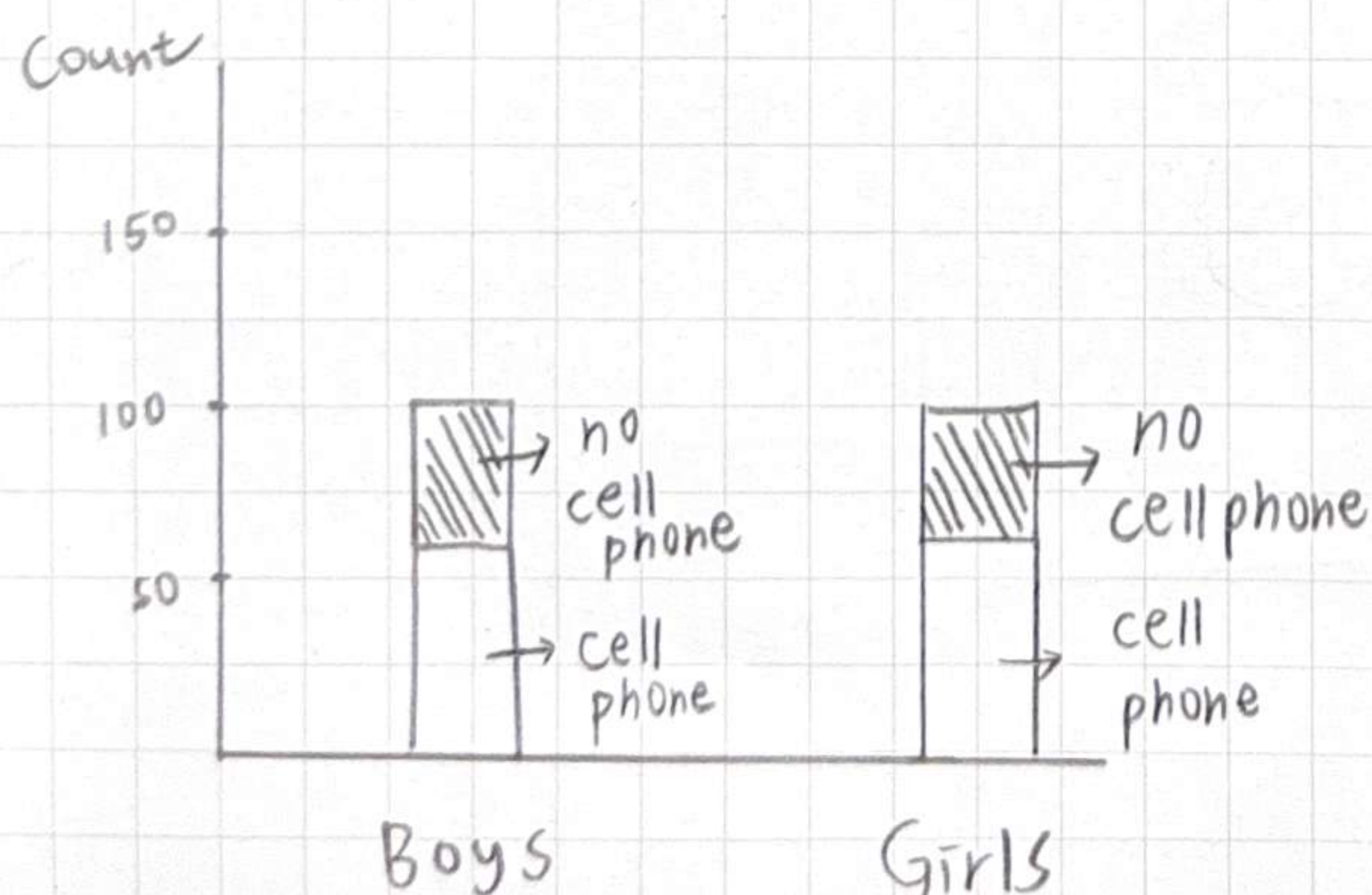
(1) Skewed to right

(2) 夏季電費較高，拉長 Box Plot 的尾端

§3.1 Q10

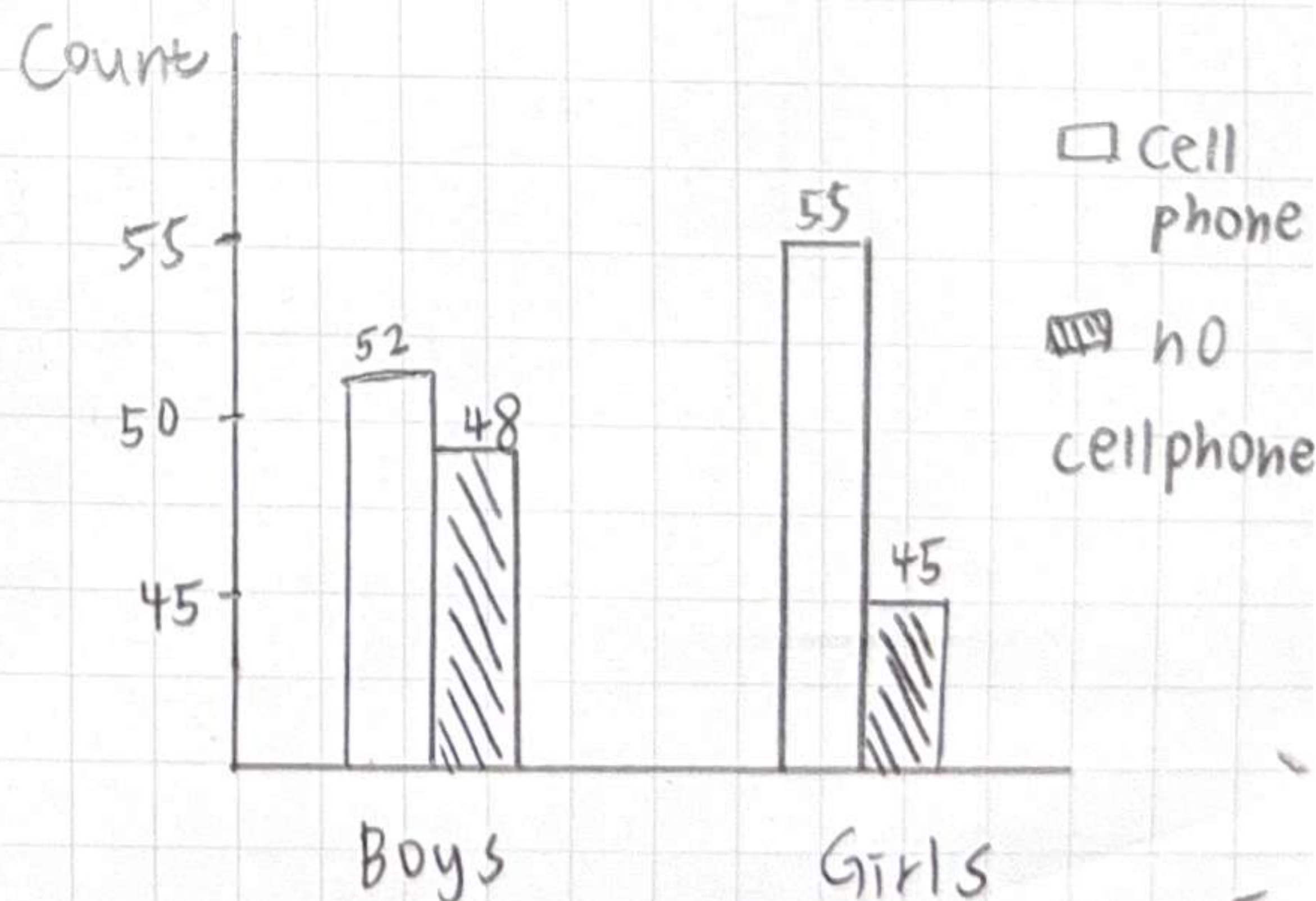
a.

Stacked bar chart



b.

side-by-side bar chart



c.

女生有手机的比例略高於男生, 但擁有手机的比列非常

相近

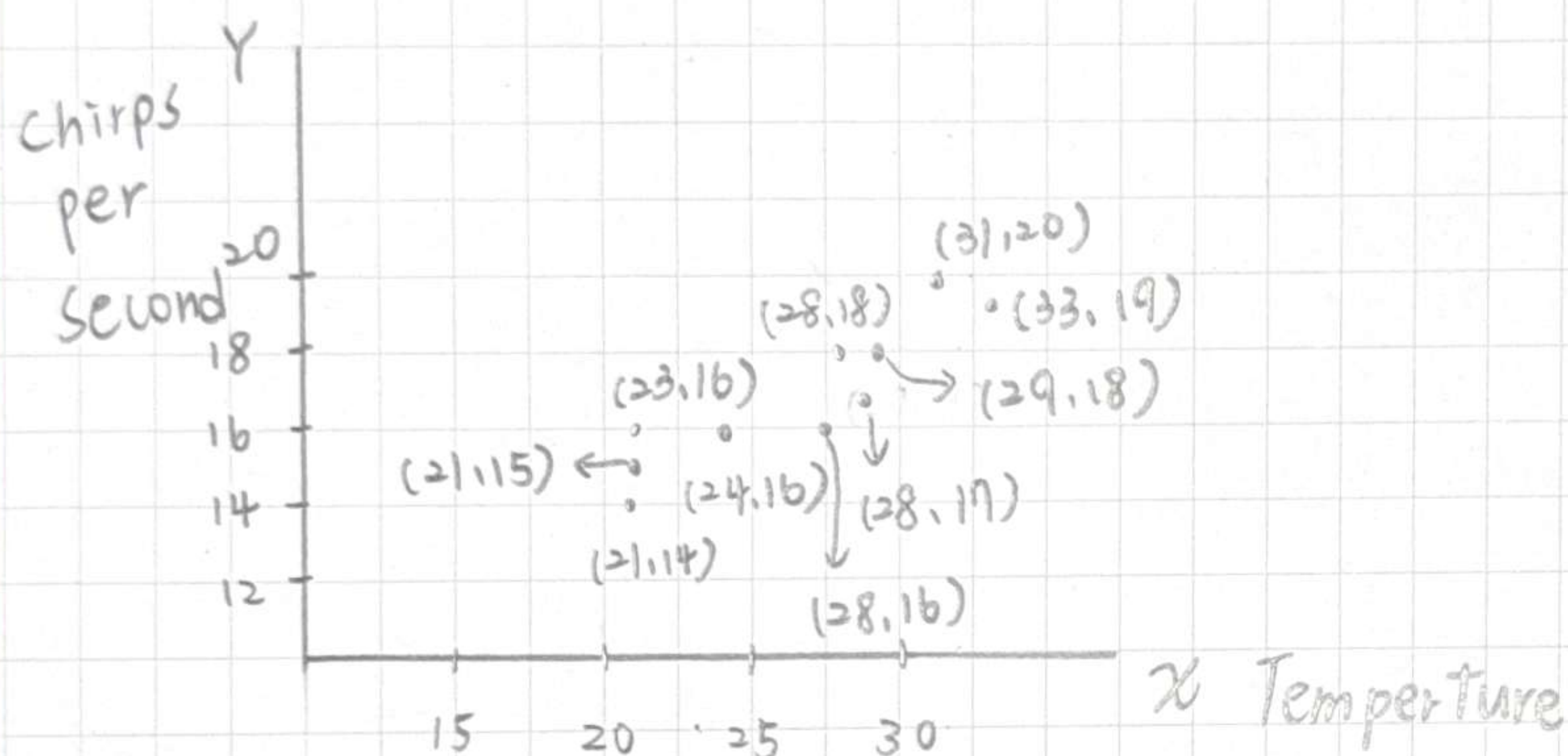
§3.2 Q25

a.

Temperature 是 independent variable (自變數)

Chirps per second 是 dependent variable (應變數)

b.



c.

$$y = bx + a$$

$$b = r \frac{S_y}{S_x}, r = \frac{S_{xy}}{S_x S_y}$$

$$a = \bar{y} - b\bar{x}$$

$$S_{xy} = \frac{4558 - \frac{266 \times 169}{10}}{9}$$

$$= 6.96$$

$$S_x^2 = \frac{7230 - \frac{(266)^2}{10}}{9}$$

$$= 17.16$$

X	Y	XY	X ²	Y ²
31	20	620	961	400
23	16	368	529	256
33	19	627	1089	361
29	18	522	841	324
28	18	504	784	324
24	16	384	576	256
21	14	294	441	196
28	17	476	784	289
21	15	315	441	225
28	16	448	784	256
SUM				
266	169	4558	7230	2887

續 § 3.2 Q25

$$s_y^2 = \frac{2887 - \frac{(169)^2}{10}}{9} = 3.43$$

$$b = r \frac{s_y}{s_x} = \frac{s_{xy}}{s_x \cdot s_x} \times \frac{s_y}{s_x} = \frac{s_{xy}}{(s_x)^2} = \frac{6.96}{17.16} = 0.406$$

$$a = \bar{y} - b\bar{x} = 16.9 - 0.406 \times 26.6 = 6.10$$

故又線性方程為 $y = 0.406x + 6.10$

d.

$$y = 0.406 \cdot 27 + 6.10 = 17.062$$

§ 3.2 Q27

a. independent variable: floors

dependent variable: height

b. 詳下頁

X	Y	XY	X ²	Y ²
73	335	24455	5329	112225
73	310	22630	5329	96100
62	262	16244	3844	68644
52	229	11908	2704	52441
52	228	11856	2704	51984
55	224	12320	3025	50176
53	221	11713	2809	48841
54	220	11880	2916	48400
52	219	11388	2704	47961
52	213	11076	2704	45369
52	213	11076	2704	45369
49	206	10094	2401	42436
54	203	10962	2916	41209
58	197	11426	3364	38809
49	194	9506	2401	37636
48	190	9120	2304	36100
42	189	7938	1764	35721
45	185	8325	2025	34225
42	176	7392	1764	30976
44	174	7656	1936	30276
44	174	7656	1936	30276
41	163	6683	1681	26569
39	162	6318	1521	26244
40	162	6480	1600	26244
40	162	6480	1600	26244
37	158	5846	1369	24964
40	157	6280	1600	24649
36	154	5544	1296	23716
SUM 1378	5680	290252	70250	1203804

$$S_{xy} = \frac{290252 - \frac{1378 \times 5680}{28}}{27} = 396.85$$

$$S_y^2 = \frac{1203804 - \frac{(5680)^2}{28}}{27} = 1910.20, S_y = 43.71$$

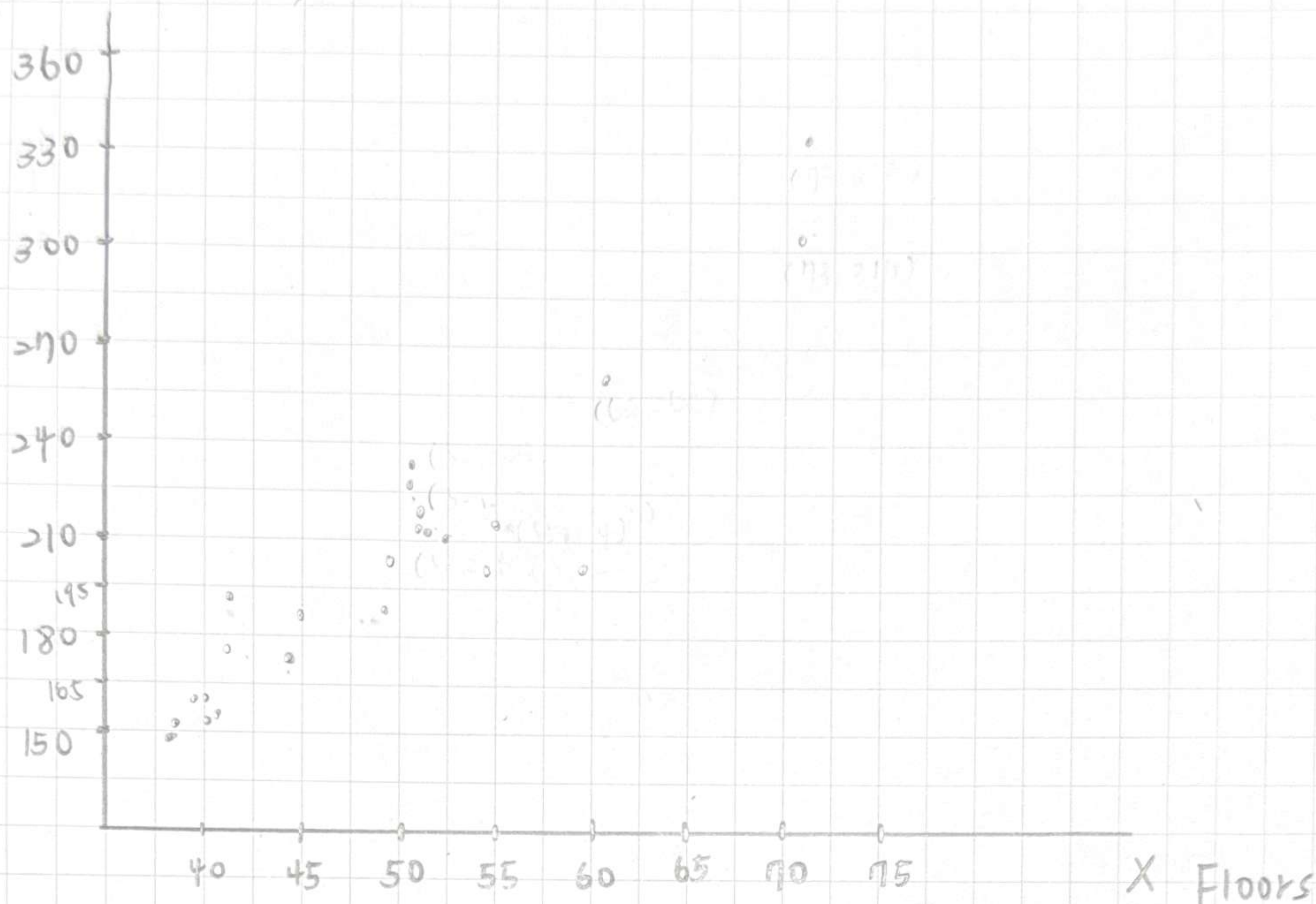
$$S_x^2 = \frac{70250 - \frac{(1378)^2}{28}}{27} = 90.10, S_x = 9.49$$

$$r = \frac{S_{xy}}{S_x \cdot S_y} = 0.957\#$$

續 §3.2 Q27

$r=0.957$, 代表高度正向線性關係, 即樓層數越多, 建築物越高

C、



正向線性關係, 無離群值

d、

$$y = bx + a$$

$$b = r \frac{s_y}{s_x} = 0.957 \times \frac{43.71}{9.49} = 4.408$$

$$a = \bar{y} - b\bar{x} = 202.86 - 4.408 \cdot 49.21 = -14.06$$

$$\Rightarrow y = 4.408x - 14.06$$

$$\text{當 } x = 48, y = 197.52 \#$$